Siddarth Madala

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EDUCATION

University of Illinois Urbana-Champaign | *PhD Computer Science* — Urbana-Champaign, IL
Advisor: Julia Hockenmaier, PhD — Aug 2022 - May 2028 (Expected)
Concentration: Natural Language Processing, Information Retrieval, Embodied AI

Duke University | *B.S. Statistics and Computer Science, Minor: Mathematics* — Aug 2018 - May 2022

EXPERIENCE

University of Illinois Urbana-Champaign — Urbana-Champaign, IL
Graduate Research Assistant — Aug 2022 - Present
- Researching information retrieval and embodied systems under Professor Julia Hockenmaier. Focused on spatio-temporal reasoning for multimodal large language models, using curriculum and reinforcement learning to improve temporal and spatial grounding.

Graduate Teaching Assistant — Aug 2022 - Present
- TA for CS 441 (Applied Machine Learning) and CS 447 (Natural Language Processing), online courses with 500+ students per semester.
- Led office hours, graded/proctored assessments, and guided student projects; translated abstract concepts into clear, actionable explanations and gave constructive feedback on design and debugging to improve student performance and comprehension.

Pinterest — New York, NY
Homefeed Ranking/Personalization Research Intern — May 2026 - Aug 2026
- Integrated PinCLIP, Pinterest's foundational vision-language model (VLM), into Homefeed's L2 late-stage ranking pipeline using RQ-VAE-derived semantic IDs, driving cross-team infrastructure and PinCorpus backfill efforts to serve 600+ million MAUs.
- Designed lightweight VLM embedding aggregation techniques that doubled training/inference throughput on large-scale distributed GPU clusters and lifted online repins by 1% at RecSys scale, sustaining 15M queries/sec with minimal latency tradeoffs in online experiments.
- Propagated experimental findings into Pinterest's flagship multi-modal Conditional Foundation Model (mCFM), extending impact across all major surfaces including Homefeed, Notifications, Board More Interests, and Search.

Epsilon — Chicago, IL
Decision Sciences/Personalization Research Intern — May 2025 - Aug 2025
- Architected a vector‑based similarity search and LLM‑driven filtering/reranking pipeline to synthesize labels for tens of millions of URLs across 200+ category tags, enabling scalable surrogate model distillation on Databricks and Apache Spark clusters.
- Engineered a hash embedding-based URL and metadata token embedding system to compress 750 GB of daily ingested URL browsing data by 70% and accelerate embedding throughput by 93x, powering both real‑time and batch advertisement personalization workflows.
- Developed and benchmarked surrogate multi-label classifiers (logistic regression, XGBoost, neural networks) on LLM‑generated synthetic labels, driving a 50% uplift in AUROC compared to legacy, hand‑tuned mapping systems.
- Scaled distributed synthetic labeling, hash-embedding and multi-label classification infrastructure to support bi‑weekly batch jobs on Databricks and local clusters, labeling and indexing trillions of URLs to feed as features in high‑throughput, real‑time ad‑bidding systems.

Elemental Cognition — New York, NY
Natural Language Processing Research Intern — May 2024 - Aug 2024
- Extended Semantic Role Labeling and Named Entity Recognition from sentence to document level by unifying 10+ corpora (PropBank, CoNLL, OntoNotes, etc.) and fine‑tuning GPT‑3.5, slashing manual annotation costs by 92% while maintaining high-quality labels.
- Deployed a real‑time token classification pipeline for document‑level SRL/NER, processing hundreds of user-uploaded documents per second to amplify LLM retrieval/reasoning and cut document query latency by 83%.

IBM Research — Yorktown Heights, NY
Hybrid Cloud Research Intern — May 2022 - Aug 2022
- Engineered a Kubernetes‑based Ray cluster pipeline on IBM Cloud to parallelize Optical Proximity Correction workloads across 20,000+ cores, boosting job throughput by 47% and cutting lithography correction times by 34%.
- Designed a containerized Electronic Design Automation (EDA) workflow platform to unify multiple research projects into a single build for testing and deployment, potentially reducing licensing costs for small chip designers by over 60%.

Quantum Computing/Machine Learning Research Intern — May 2021 - Aug 2021
- Pioneered a novel Qiskit‑based quantum twin generator to model digital circuits in the quantum domain, delivering logic verification speeds 200% faster than traditional boolean EDA workflows.
- Developed a graph neural network system leveraging macro and connectivity data from intermediate EDA logs to predict micro-processor die size, cutting design iteration cycles from weeks to days.

RESEARCH

- Lee, J. et al. “ReasoningFlow: Discourse Structures for Understanding LLM Reasoning Traces.” *EMNLP 2026.* arXiv:2606.05402
- Huang, J. et al. “Contextual Relevance and Adaptive Sampling for LLM-Based Document Reranking.” *ACL 2026.* 2026.acl-long.94
- Huang, J. et al. “Tackling Distractor Documents in Multi-Hop QA with Reinforcement and Curriculum Learning.” *EACL Findings 2026.* 2026.findings-eacl.294

Skills: Python, Java, C, C++, Rust, R, Bash, PostgreSQL, PyTorch, Git, Databricks, Spark, Hadoop, AWS, Docker/Kubernetes, Figma